IT TECHNOLOGY SKILLS DEMAND

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OUTLINE

- Executive Summary
- Introduction
- Methodology
- Results
  - Visualization – Charts
  - Dashboard
- Discussion
  - Findings & Implications
- Conclusion
- Appendix

EXECUTIVE SUMMARY

- This report analyzes global developer data to identify the most in-demand programming skills.
- Data was collected from job postings, web scraping, and developer survey datasets.
- The analysis reveals that a small group of programming languages and databases dominate the current market.
- At the same time, there is a growing interest in emerging technologies for future use.
- The findings highlight the importance of continuous learning and adapting to evolving industry trends.

Key Points:

1: Core technologies remain highly in demand

2: Future trends show a shift toward modern tools

- Growth in new programming languages
- Increased interest in scalable databases
- Adoption of modern development tools

3: Data-driven skills are becoming essential

4: Developers must continuously upskill

5: Organizations should align training with trends

INTRODUCTION

- The purpose of this report is to analyze current and future technology skill demands using real-world data.
- The target audience includes business leaders, HR professionals, and developers seeking to understand industry trends.
- This analysis provides valuable insights to support hiring, training, and strategic planning decisions.

Key Points:

1: Rapid evolution of the technology landscape

2: Increasing demand for multi-skilled professionals

3: Importance of identifying emerging skills

4: Role of data analysis in decision-making

- Use of multiple datasets
- Data-driven insights for strategy

METHODOLOGY

- Data was collected from multiple sources including job postings APIs, web scraping, and developer surveys.
- The datasets were processed using Python for cleaning, transformation, and analysis.
- Data wrangling included handling missing values, removing duplicates, and standardizing formats.
- Statistical analysis and visualizations were created to identify trends.

Key Points:

1: Data collection using APIs and web scraping

2: Data cleaning and preprocessing

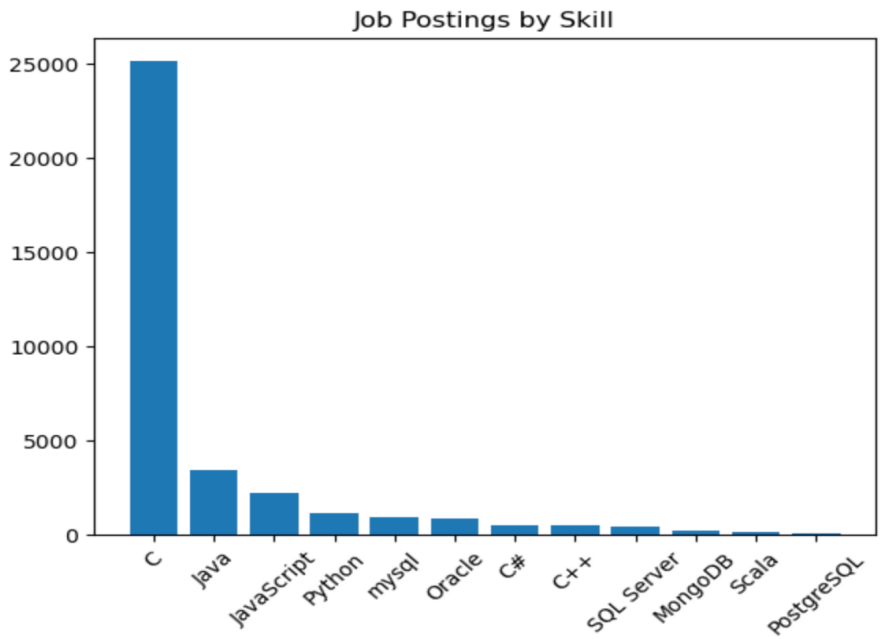
3: Data transformation and aggregation

4: Visualization using charts and dashboards

- Bar charts for comparison
- Dashboard for insights

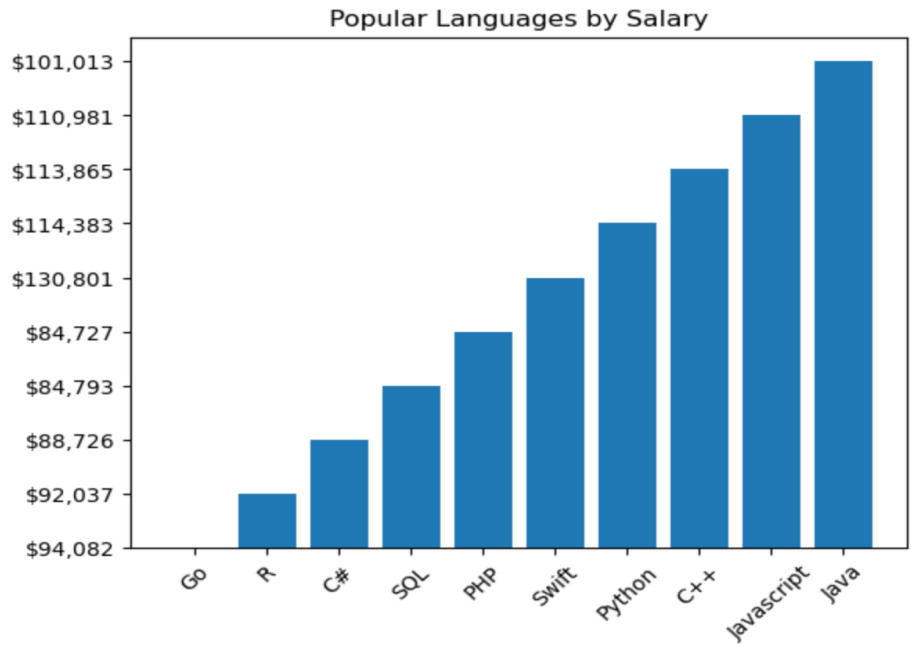
Results

JOB POSTINGS



The chart shows the distribution of job postings across different programming languages, highlighting the most in-demand skills in the job market.

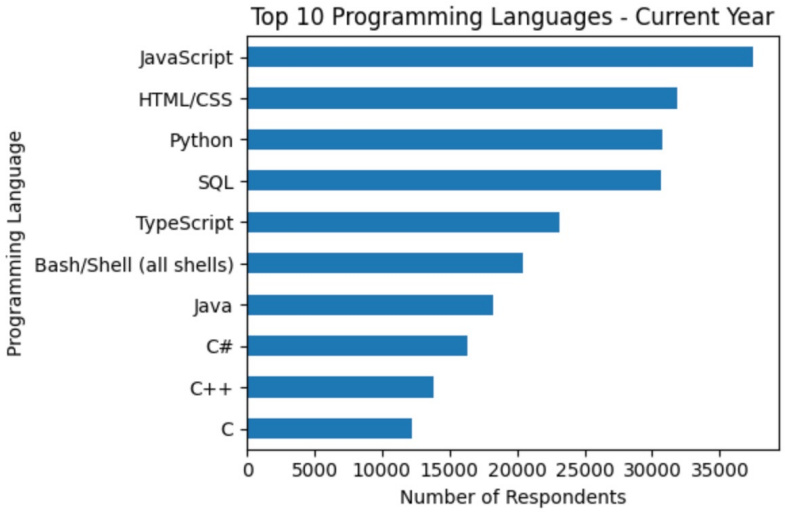
POPULAR LANGUAGES



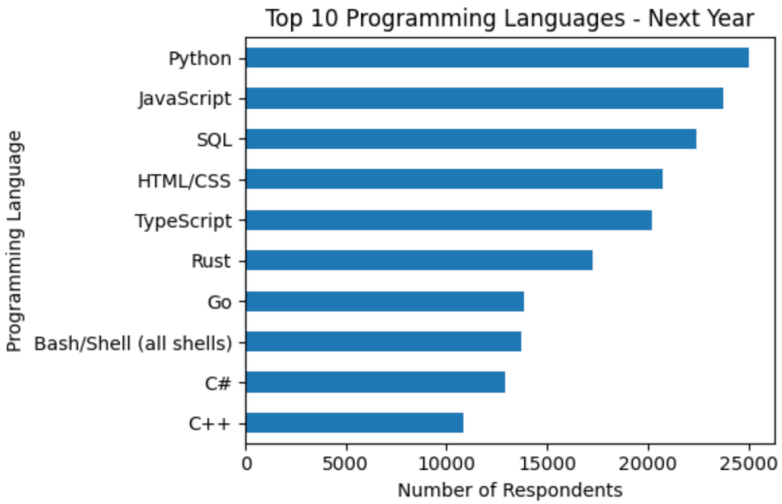
The chart presents programming languages ranked by average salary, indicating the highest-paying technologies.

PROGRAMMING LANGUAGE TRENDS

Current Year



Next Year



SUMMARY: The analysis shows that a small group of programming languages dominates both current usage and future demand. Languages used in the current year remain highly relevant in the next year, indicating strong consistency in industry demand. Emerging technologies are gradually gaining attention, but core languages continue to lead.

PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

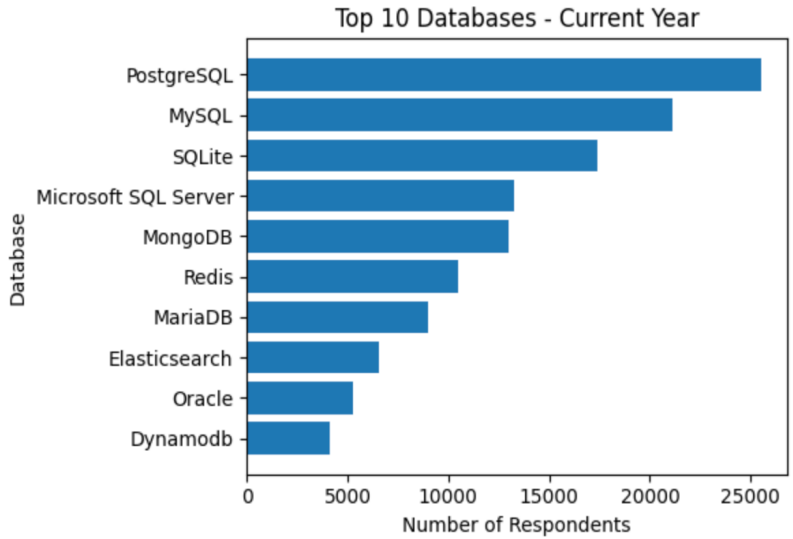
1. A few programming languages account for the majority of usage among developers.
2. There is strong overlap between current usage and future preferences.
3. Newer languages are growing but have not yet overtaken established ones

Implications

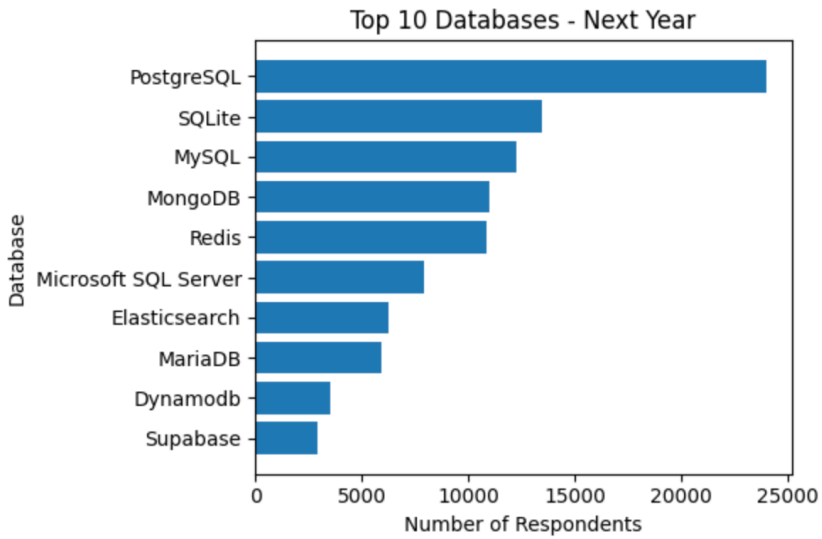
1. Focus on mastering widely used languages for better job opportunities.
2. Continuous learning is important to keep up with evolving trends.
3. Organizations should prioritize training in high-demand core technologies.

DATABASE TRENDS

Current Year



Next Year



SUMMARY: Database technologies show a strong preference for relational databases in current usage. At the same time, there is a noticeable increase in interest toward modern and NoSQL databases for future use. This reflects a shift toward handling large-scale and unstructured data.

DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

1. Traditional relational databases dominate current usage.
2. Modern databases are gaining popularity among developers.
3. There is a clear transition toward scalable and flexible database solutions.

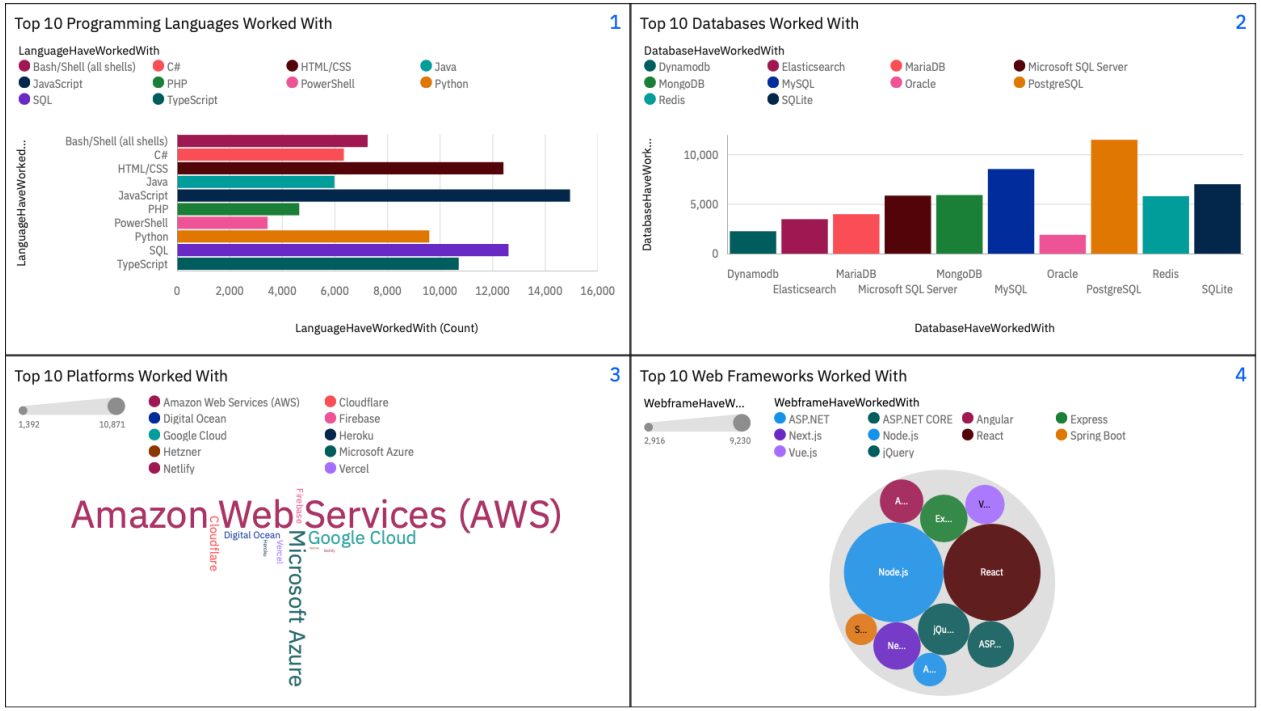
Implications

1. Strong SQL knowledge remains essential for developers.
2. Learning NoSQL and cloud-based databases will provide a competitive advantage.
3. Organizations should adopt hybrid database strategies.

DASHBOARD

DASHBOARD TAB 1: Current Technology Usage

CURRENT TECHNOLOGY USAGE



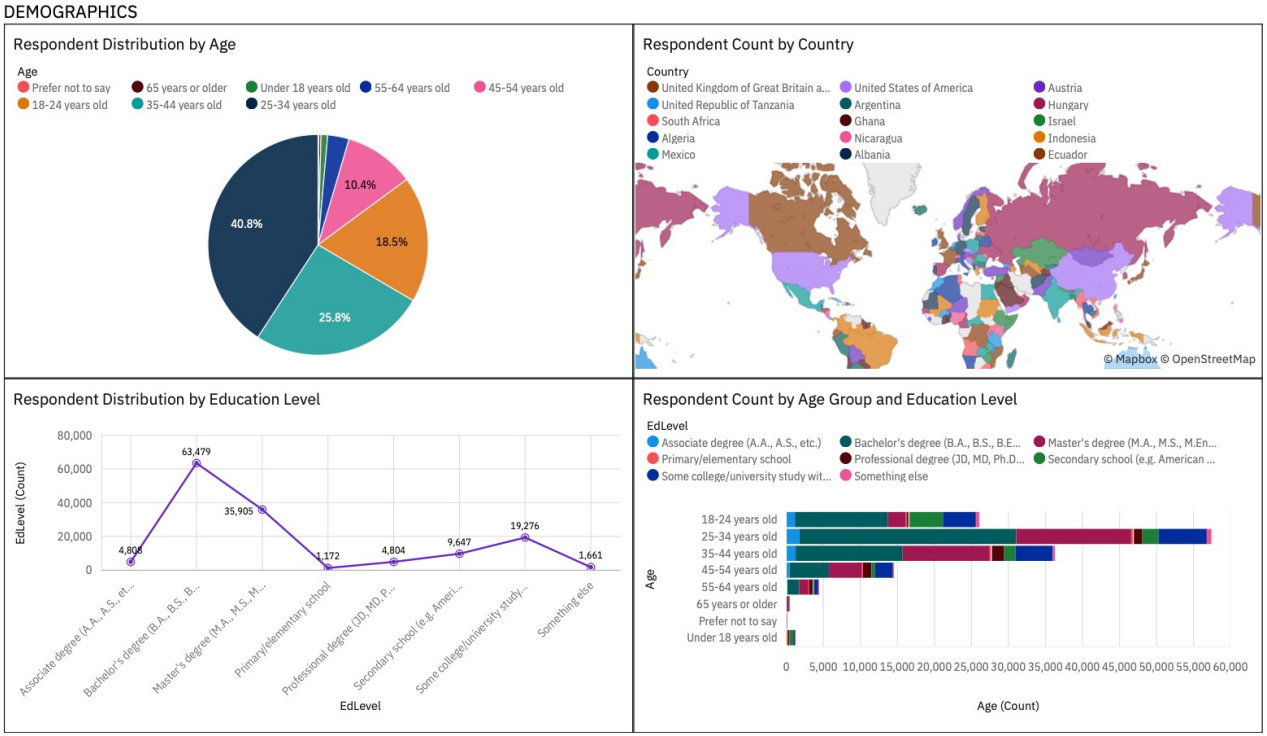
1. The dashboard highlights that a few core technologies dominate current usage across developers.
2. There is high concentration in specific programming languages, databases, and tools.
3. This indicates a relatively stable technology stack in the present industry.

DASHBOARD TAB 2: Future Technology Trends

FUTURE TECHNOLOGY TREND

1. The dashboard shows a shift toward newer and emerging technologies.
2. Developers are increasingly interested in modern tools and frameworks.
3. This suggests a transition toward innovation-driven skill demand.

DASHBOARD TAB 3: Demographics



1. The developer population is diverse, with representation across different experience levels and regions.
2. A large proportion of developers are early-career professionals.
3. This indicates a growing and evolving global developer ecosystem.

DISCUSSION

1. The analysis demonstrates that while core technologies remain dominant, the industry is gradually evolving toward modern and scalable solutions.
2. Developers are balancing between maintaining foundational skills and exploring new technologies.
3. This trend highlights the importance of adaptability and continuous learning.

OVERALL FINDINGS & IMPLICATIONS

Findings

1. Core programming and database technologies remain dominant.
2. Future trends indicate a shift toward emerging technologies.
3. Skill demand is evolving but still rooted in foundational tools.

Implications

1. Developers should focus on both core and emerging skills.
2. Organizations must invest in upskilling programs.
3. Data-driven insights are essential for workforce planning.

CONCLUSION

- ✓ The demand for technology skills continues to evolve rapidly across industries.
- ✓ Core programming languages and databases remain essential for most roles.
- ✓ Future demand is shifting toward modern, scalable, and data-driven technologies.
- ✓ Continuous learning and adaptability are critical for long-term career success.

APPENDIX

- Sample datasets used for analysis (job postings, survey data, web-scraped data)
- Additional charts created during analysis
- Data preprocessing steps including splitting and cleaning multi-value fields
- Tools used: Python (Pandas, Matplotlib), Excel, IBM Cognos Analytics